RACE Regular Analysis Through Continuous Exercise

Binomial Theorem: RACE-01

Single Correct Type Questions:

1. 16th term in the expansion of $(\sqrt{x}-\sqrt{y})^{17}$ is

(A) $136xy^7$ (B) $136xy$ (C) $-136xy^{15/2}$ (D) $-136xy^2$

MBT167

2. Coefficient of x in the expansion of $\left(x^2+\frac{a}{x}\right)^5$ is

(A) $9a^2$ (B) $10a^3$ (C) $10a^2$ (D) $10a$

MBT168

3. The coefficient of x^7 in the expansion of $\left(\frac{x^2}{2}-\frac{2}{x}\right)^8$ is

(A) –56 (B) 56 (C) –14 (D) 14

MBT169

4. In the expansion of $\left(x-\frac{1}{x}\right)^6$, the term independent of x is

(A) –20 (B) 20 (C) 30 (D) –30

MBT170

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5. The coefficient of x^{32} in the expansion of $\left(x^4-\frac{1}{x^3}\right)^{15}$ is

(A) ${}^{15}C_5$ (B) ${}^{15}C_6$ (C) ${}^{15}C_4$ (D) ${}^{15}C_7$

MBT171

6. The term independent of x in the expansion of $\left(\frac{1}{2}x^{1/3}+x^{-1/5}\right)^8$ will be

(A) 5 (B) 6 (C) 7 (D) 8

MBT172

7. In the expansion of $(1+x+x^3+x^4)^{10}$ the coefficient of x^4 is

(A) ${}^{40}C_4$ (B) ${}^{10}C_4$ (C) 210 (D) 310

MBT173

8. The coefficient of middle term in the expansion of $(1+x)^{10}$ is

(A) $\frac{10!}{5!6!}$ (B) $\frac{10!}{(5!)^2}$ (C) $\frac{10!}{5!7!}$ (D) None of these

MBT174

9. The number of dissimilar terms in the expansion of $(1+x^4-2x^2)^{15}$ is

(A) 21 (B) 31 (C) 41 (D) 61

MBT175

For More Join: @IITJEE_Advanced

10. Find the middle term(s) in the expansion of $\left(2y-\frac{y^2}{2}\right)^{11}$

(A) $-2\,{}^{11}C_5 y^{16}$ & $\frac{{}^{11}C_6}{2}y^{17}$ (B) $2\,{}^{11}C_5 y^{16}$ & $\frac{{}^{11}C_6}{2}y^{17}$

(C) $2\,{}^{11}C_5 y^{16}$ & $-\frac{{}^{11}C_6}{2}y^{17}$ (D) $-2\,{}^{11}C_5 y^{16}$ & $-\frac{{}^{11}C_6}{2}y^{17}$

MBT176

11. The value of $\frac{18^3+7^3+3.18.7.25}{3^6+6.243.2+15.81.4+20.27.8+15.9.16+6.3.32+64}$ is :

(A) 1 (B) 2 (C) 3 (D) None

MBT177

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	C	B	C	A	C	C	D	B	B	A	A

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Binomial Theorem: RACE-02

Single Correct Type Questions:

1. ${}^{15}C_{3r} = {}^{15}C_{r+3}$ then value of r is

(A) 3 (B) 4

(C) 5 (D) 8

MBT178

2. The value of ${}^{15}C_3 + {}^{15}C_{13}$ is

(A) ${}^{16}C_3$ (B) ${}^{30}C_{16}$

(C) ${}^{15}C_{10}$ (D) ${}^{16}C_{14}$

MBT179

3. If $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \ldots + {}^nC_nx^n$, then the value of ${}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n$ is -

(A) 2^{n+1} (B) 2^{n-1}

(C) $2^n + 1$ (D) $2^n - 1$

MBT180

4. If $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \ldots + {}^nC_n.x^n$ then the value of ${}^nC_0 + 3\cdot{}^nC_1 + 5\cdot{}^nC_2 + \ldots + (2n+1)\,{}^nC_n$ is -

(A) $n.2^n$ (B) $(n-1).2^n$

(C) $(n+2).2^{n-1}$ (D) $(n+1).2^n$

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MBT181

5. The value of $\sum_{r=1}^{10} r.\frac{{}^nC_r}{{}^nC_{r-1}}$ is equal to :

(A) $5(2n-9)$ (B) $10n$

(C) $9(n-4)$ (D) $n-4$

MBT182

6. If $(1+x)^n = \sum_{r=0}^{n} a_r x^r, b_r = 1 + \frac{a_r}{a_r - 1}$ and $\prod_{r=1}^{n} b_r = \frac{(101)^{100}}{100!}$, then n equals to :

(A) 99 (B) 100

(C) 101 (D) 102

MBT183

7. $\sum_{r=0}^{n-1} \frac{{}^nC_r}{{}^nC_r + {}^nC_{r+1}} =$

(A) $\frac{n}{2}$ (B) $\frac{n+1}{2}$

(C) $(n+1)\frac{n}{2}$ (D) $\frac{n(n-1)}{2(n+1)}$

MBT184

For More Join: @IITJEE_Advanced

8. $\frac{^{11}C_0}{1}+\frac{^{11}C_1}{2}+\frac{^{11}C_2}{3}+\ldots\ldots+\frac{^{11}C_{10}}{11}=$

(A) $\frac{2^{11}-1}{11}$ (B) $\frac{2^{11}-1}{6}$

(C) $\frac{3^{11}-1}{11}$ (D) $\frac{3^{11}-1}{6}$

MBT185

9. If the sum of the coefficients in the expansion of $(2+3cx+c^2x^2)^{12}$ vanishes, then c equals to

(A) –1, 2 (B) 1, 2

(C) 1, –2 (D) –1, –2

MBT186

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	A	A	D	D	A	B	A	B	D

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Binomial Theorem: RACE-03

Single Correct Type Questions:

1. The remainder when 17^{10} is divided by 9 is:
(A) 8 (B) 7 (C) 2 (D) 1
MBT187

2. The remainder when 7^{98} is divided by 5 is
(A) 4 (B) 0 (C) 2 (D) 3
MBT188

3. The remainder when 2^{2003} is divided by 17 is :
(A) 1 (B) 2 (C) 8 (D) none of these
MBT189

4. The last three digits of 3^{50} will be
(A) 249 (B) 259 (C) 349 (D) 241
MBT190

5. The last two digits of the number 3^{400} is :
(A) 81 (B) 43 (C) 29 (D) 01
MBT192

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6. If $n \in N$ then $10^n + 5$ is divisible by :
(A) 3 (B) 7 (C) 9 (D) 2
MBT193

7. If the numerically greatest term in the expansion of $(3-5x)^{15}$ when $x = \frac{1}{5}$ is $455 \times 3^n, n \in N$ then value of n is
(A) 10 (B) 12 (C) 14 (D) 16
MBT194

8. The greatest term in the expansion of $\sqrt{3}\left(1+\frac{1}{\sqrt{3}}\right)^{20}$ is
(A) $\frac{25840}{9}$ (B) $\frac{24840}{9}$ (C) $\frac{26840}{9}$ (D) None of these
MBT195

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	D	A	C	A	D	A	B	A

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Binomial Theorem: RACE-04

Single Correct Type Questions:

1. The sum of the coefficients of even powers of x in the expansion of $(1+x+x^2+x^3)^5$ is -

(A) 512 (B) –512 (C) 215 (D) none of these

MBT196

2. The value of the expression $\left(\sum_{r=0}^{10} {}^{10}C_r\right)\left(\sum_{K=0}^{10}(-1)^K \frac{{}^{10}C_K}{2^K}\right)$ is :

(A) 2^{10} (B) 2^{20} (C) 1 (D) 2^5

MBT197

3. The value of $\binom{50}{0}\binom{50}{1}+\binom{50}{1}\binom{50}{2}+\ldots\ldots+\binom{50}{49}\binom{50}{50}$ is, where ${}^nC_r=\binom{n}{r}$

(A) $\binom{100}{50}$ (B) $\binom{100}{51}$ (C) $\binom{50}{25}$ (D) $\binom{50}{25}^2$

MBT198

4. In the expansion of $(1+x)^n\left(1+\frac{1}{x}\right)^n$, the term independent of x is -

(A) $C_0^2+2C_1^2+\ldots+(n+1)C_n^2$ (B) $(C_0+C_1+\ldots+C_n)^2$

(C) $C_0^2+C_1^2+\ldots+C_n^2$ (D) None of these

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MBT199

5. If $(1+x)^n=C_0+C_1x+C_2x^2+\ldots+C_nx^n$, then find the value of $C_0^2+C_1^2+C_2^2+\ldots+C_n^2=$

(A) ${}^nC_n$ (B) ${}^{2n}C_n$ (C) ${}^{3n}C_n$ (D) ${}^{4n}C_n$

MBT200

6. If $(1+x)^n=C_0+C_1x+C_2x^2+\ldots+C_nx^n$, then find the value of

$C_0C_2+C_1C_3+C_2C_4+\ldots+C_{n-2}C_n=$

(A) ${}^{2n}C_{n-3}$ (B) ${}^nC_{n-2}$ (C) ${}^{2n}C_{n-2}$ (D) ${}^{2n}C_{n-4}$

MBT201

7. If $|x|<1$ then 7th term in the expansion of $(1-x)^{-2}$ is

(A) $7x^7$ (B) $6x^7$ (C) $7x^6$ (D) $6x^6$

MBT202

8. $\left(x^2+\frac{1}{x}\right)^{-4/3}$ can be expanded by binomial theorem if

(A) $x<1$ (B) $|x|<1$ (C) $x>1$ (D) $|x|>1$

MBT203

9. If $|x|<1$, then the co-efficient of x^n in the expansion of $(1+x+x^2+x^3+\ldots\ldots)^2$ is

(A) n (B) $n-1$ (C) $n+2$ (D) $n+1$

MBT204

For More Join: @IITJEE_Advanced

10. The coefficient of x^4 in the expression $(1 + 2x + 3x^2 + 4x^3 + \ldots\ldots upto\ \infty)^{\frac{1}{2}}$ (where $|x| < 1$) is

(A) 1 (B) 3 (C) 2 (D) 5

MBT205

Multiple Correct Type Questions:

11. The coefficient of x^4 in $\left(\frac{1+x}{1-x}\right)^2$, $|x| < 1$, is

(A) 4 (B) – 4 (C) $10 + {}^4C_2$ (D) 16

MBT206

12. The expansion of $(3x + 2)^{-1/2}$ is valid in ascending powers of x, if x lies in the interval.

(A) $(0, 2/3)$ (B) $(-3/2, 3/2)$ (C) $(-2/3, 2/3)$ (D) $(-\infty, -3/2)\ (3/2, \infty)$

MBT207

13. If $\left(3\sqrt{3} + 5\right)^n = p + f$, where p is an integer and f is a proper fraction, then find the value of $\left(3\sqrt{3} - 5\right)^n$, $n \in N$, is

(A) $1 - f$, if n is even (B) f, if n is even (C) $1 - f$, if n is odd (D) f, if n is odd

MBT208

14. If $\left(9 + \sqrt{80}\right)^n = I + f$, where I, n are integers and $0 < f < 1$, then:

(A) I is an odd integer (B) I is an even integer

(C) $(I + f)(1 - f) = 1$ (D) $1 - f = \left(9 - \sqrt{80}\right)^n$

MBT209

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	B	C	B	C	C	D	D	A
Question	11	12	13	14						
Answer	C,D	A,C	A,D	A,C,D						

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Binomial Theorem: RACE-05

Single Correct Type Questions:

1. The value of $\binom{30}{0}\binom{30}{10}-\binom{30}{1}\binom{30}{11}+\binom{30}{2}\binom{30}{12}\ldots+\binom{30}{20}\binom{30}{30}$, is where $\binom{n}{r}={}^{n}C_{r}$

(A) ${}^{30}C_{10}$ (B) ${}^{60}C_{20}$

(C) ${}^{31}C_{11}$ or ${}^{31}C_{10}$ (D) ${}^{30}C_{11}$

MBT210

2. For r =0, 1,....,10, let A_r, B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}, (1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r(B_{10}B_r - C_{10}A_r)$ is equal to :

(A) $B_{10} - C_{10}$ (B) $A_{10}(B_{10}^2 - C_{10}A_{10})$

(C) 0 (D) $C_{10} - B_{10}$

MBT211

3. Coefficient of x^{11} in the expansion of $(1+x^2)^4(1+x^3)^7(1+x^4)^{12}$ is -

(A) 1051 (B) 1106

(C) 1113 (D) 1120

MBT212

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4. The sum of coefficients of integral powers of x in the binomial expansion of $(1-2\sqrt{x})^{50}$ is :

(A) $\frac{1}{2}\left(3^{50}-1\right)$ (B) $\frac{1}{2}\left(2^{50}+1\right)$

(C) $\frac{1}{2}\left(3^{50}+1\right)$ (D) $\frac{1}{2}\left(3^{50}\right)$

MBT213

5. Statement –1 : $\sum_{r=0}^{n}(r+1)\,{}^{n}C_{r} = (n+2)2^{n-1}$

Statement – 2 : $\sum_{r=0}^{n}(r+1)\,{}^{n}C_{r}\,x^{r} = (1+x)^{n} + nx(1+x)^{n-1}$

(A) Statement –1 is false, Statement –2 is true

(B) Statement–1 is true, Statement–2 is true; Statement–2 is a correct explanation for Statement–1

(C) Statement–1 is true, Statement–2 is true; Statement–2 is not a correct explanation for Statement–1

(D) Statement–1 is true, Statement–2 is false

MBT214

6. The remainder left out when $8^{2^n} - (62)^{2^{n+1}}$ is divided by 9 is :

(A) 7 (B) 8

(C) 0 (D) 2

MBT215

For More Join: @IITJEE_Advanced

7. Let $S_1 = \sum_{j=1}^{10} j(j-1)\ {}^{10}C_j$, $S_2 = \sum_{j=1}^{10} j\,{}^{10}C_j$ and $S_3 = \sum_{j=1}^{10} j^2\,{}^{10}C_j$

Statement–1 : $S_3 = 55 \times 2^9$.

Statement–2 : $S_1 = 90 \times 2^8$ and $S_2 = 10 \times 2^8$.

(A) Statement–1 is true, Statement–2 is true; Statement–2 is a correct explanation for Statement–1.

(B) Statement–1 is true, Statement–2 is true; Statement–2 is not a correct explanation for statement–1.

(C) Statement–1 is true, Statement–2 is false.

(D) Statement–1 is false, Statement–2 is true.

MBT216

8. The coefficient of x^7 in the expansion of $(1-x-x^2+x^3)^6$ is :

(A) –144 (B) 132 (C) 144 (D) –132

MBT217

9. If n is a positive integer, then $\left(\sqrt{3}+1\right)^{2n} - \left(\sqrt{3}-1\right)^{2n}$ is :

(A) a rational number other than positive integers

(B) an irrational number

(C) an odd positive integer

(D) an even positive integer

MBT218

10. The coefficients of three consecutive terms of $(1+x)^{n+5}$ are in the ratio 5: 10: 14. Then n =

(A) 5 (B) 6 (C) 7 (D) 8

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MBT219

11. $\sum_{r=0}^{n} (-1)^r\, {}^nC_r \cdot \frac{(1+r\ell n10)}{(1+\ell n10^n)^r} =$

(A) 0 (B) ½ (C) 1 (D) None of these

MBT220

12. Coefficient of x^{n-1} in the expansion of,

$(x+3)^n + (x+3)^{n-1}(x+2) + (x+3)^{n-2}(x+2)^2 + \ldots\ldots + (x+2)^n$ is :

(A) ${}^{n+1}C_2(3)$ (B) ${}^{n-1}C_2(5)$

(C) ${}^{n+1}C_2(5)$ (D) ${}^{n}C_2(5)$

MBT221

13. If $(1+x)^n = \sum_{r=0}^{n} a_r x^r$ and $b_r = 1 + \frac{a_r}{a_{r-1}}$ and $\prod_{r=1}^{n} b_r = \frac{(101)^{100}}{100!}$, then n equals to :

(A) 99 (B) 100 (C) 101 (D) 102

MBT222

14. Number of rational terms in the expansion of $\left(1+\sqrt{2}+\sqrt{5}\right)^6$ is :

(A) 7 (B) 10 (C) 6 (D) 8

MBT223

15. If $S = {}^{404}C_4 - {}^4C_1 \cdot {}^{303}C_4 + {}^4C_2 \cdot {}^{202}C_4 - {}^4C_3 \cdot {}^{101}C_4 = (101)^k$ then k equals to :

(A) 1 (B) 2 (C) 4 (D) 6

MBT224

For More Join: @IITJEE_Advanced

Multiple Correct type Questions:

16. In the expansion of $(x + y + z)^{25}$

(A) every term is of the form ${}^{25}C_r . {}^{r}C_k . x^{25-r} . y^{r-k} . z^k$

(B) the coefficient of $x^8y^9z^9$ is 0

(C) the number of terms is 325

(D) none of these

MBT225

17. If $(1 + x + 2x^2)^{20} = a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{40}x^{40}$, then $a_0 + a_2 + a_4 \ldots\ldots + a_{38}$ is equal to

(A) $2^{19}(2^{30} + 1)$ (B) $2^{19}(2^{20} - 1)$

(C) $2^{39} - 2^{19}$ (D) $2^{39} + 2^{19}$

MBT226

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	D	C	C	B	C	C	A	B	B
Question	11	12	13	14	15	16	17			
Answer	A	C	B	B	C	A,B	B,C			

For More Join: @IITJEE_Advanced